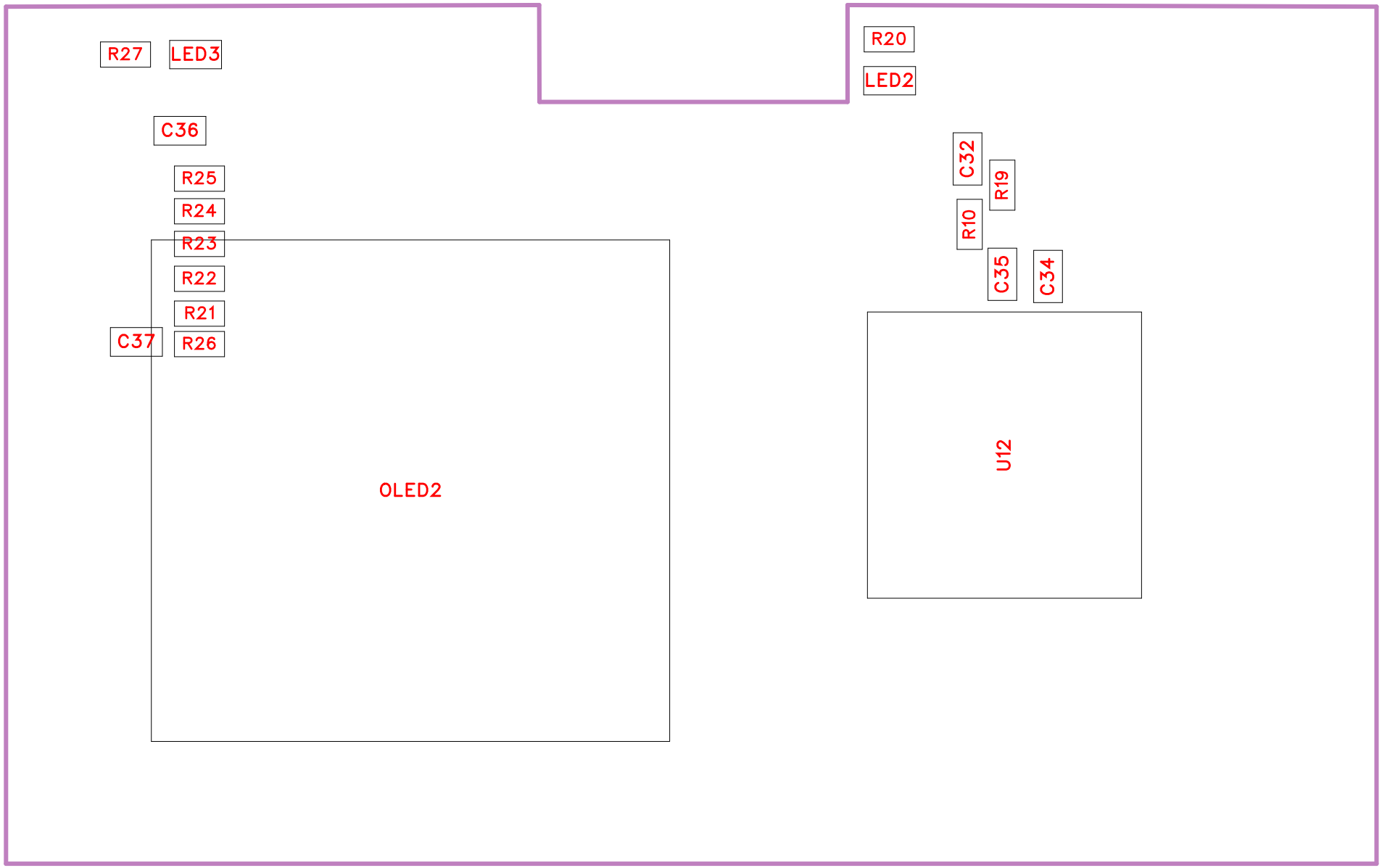
SW1
LED1
R17
LED4
D2
R49
C48
C27
C28
U15
D1
VCC
OUT1
C29
L11
BAT
DAC1
C47
U21
R13
R14
CARD1
R15
R11
U6
U5
R16
MSD1
SD1
R12
USB1
R8
U3
R9
C33
TXD0
RXD0
R51
C49
R52
R53
IO0
EN1
U17
C39
Q1
R29
C52
U25
C53
Q2
R28
R5
U4
C2
L5
R6
R4
C3
R3
C46
L8
L6
C43
L4
C5
L3
C4
U1
R2
C6
R50
U2
U23
L7
R1
L12
U24
C42
C41
L2
L1
C8
C7
C44
C45

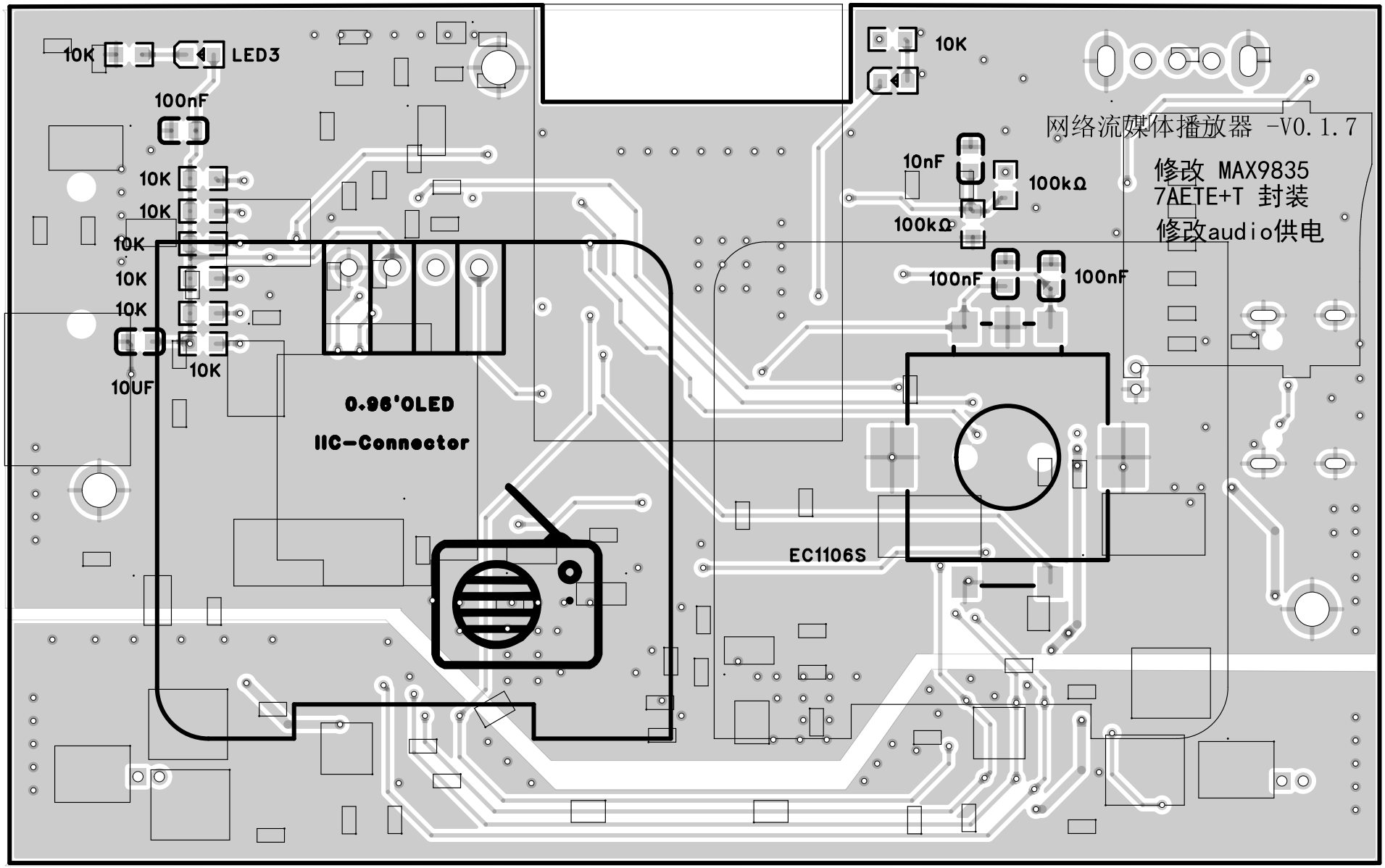

R27
LED3
C36
R25
R24
R23
R22
R21
C37
R26
OLED2
R20
LED2
C32
R19
R10
C35
C34
U12

No.	Category	Quantity	Designator	Footprint	Value	Manufacturer Part	JLCPCB Price	Comment	LCSC Price	Device	Name
1		6	BAT_DAC1,MISO1,RXD0,SCL1,TXD0,VCC_OUT1	Test-Point-0.5mm				Test-Point		Test-Point	
2		2	C2,C5	C0603_4	100nF	CL10B104KO8NNNC	0.0265	100nF	0.0265	CL10B104KO8NNNC	100nF
3		2	C3,C6	C0603_4	10uF	CL10A106MQ8NNNC	0.038	10uF	0.038	CL10A106MQ8NNNC	10uF
4		1	C4	C0603_4	22pF	CL10C220JB8NNNC	0.0238	22pF	0.0238	CL10C220JB8NNNC	22pF
5	Capacitors:Multilayer Ceramic Capacitors MLCC - SMD/SMT	2	C7,C41	C0603_5	1uF	0603F105M160NT	0.0326	1uF	0.0326	0603F105M160NT	
6	Capacitors:Multilayer Ceramic Capacitors MLCC - SMD/SMT	2	C8,C42	C0603_5	10uF	GRM188R6YA106MA73J	0.705	10uF	0.7047	GRM188R6YA106MA73J	
7	Capacitors:Multilayer Ceramic Capacitors MLCC - SMD/SMT	3	C27,C29,C52	C0603	10uF	CL10A106MA8NRNC	0.113	10uF	0.1128	CL10A106MA8NRNC	10uF
8	Capacitors:Multilayer Ceramic Capacitors MLCC - SMD/SMT	2	C28,C53	c0603_1	10uF	CL10A106MA8NRNC	0.113	10uF	0.1128	CL10A106MA8NRNC	10uF
9	Capacitors:Multilayer Ceramic Capacitors MLCC - SMD/SMT	1	C32	C0603	10nF	CL10B103KB8NNNC	0.022	10nF	0.022	CL10B103KB8NNNC	10nF
10	Capacitors:Multilayer Ceramic Capacitors MLCC - SMD/SMT	3	C33,C34,C35	c0603_1	100nF	CL10B104KB8NNNC	0.0175	100nF	0.0175	CL10B104KB8NNNC	100nF
11		3	C36,C47,C48	C0603_4	100nF			100nF		CAP_0603	100nF
12		1	C37	C0603_4	10UF			10UF		CAP_0603	10UF
13		1	C39	C0603_4	100nF/50v			100nF/50v		CAP_0603	100nF/50v

14	Capacitors:Multilayer Ceramic Capacitors MLCC - SMD/SMT	4	C43,C44,C45,C46	C0603_5	220nF	CL10B224KA8NNNC	0.0468	220nF	0.0468	CL10B224KA8NNNC	
15		1	C49	C0603_3				10uF		10uF	10uF
16	连接器:SD卡/存储卡连接器	1	CARD1	TF-SMD_TF-PUSH		TF PUSH	0.401	TF PUSH	0.4041	TF PUSH	TF PUSH
17	二极管:肖特基二极管	1	D1	SMA_L4.3-W2.6-LS5.2-RD		SS34	0.186	SS34	0.1857	SS34_C8678	SS34
18	TVS/保险丝/板级保护:静电放电(ESD)保护器件	1	D2	SOD-323_L1.7-W1.3-LS2.5-BI		TPPESD3V3L1BA	0.283	TPPESD3V3L1BA	0.2829	TPPESD3V3L1BA	TPPESD3V3L1BA
19	Key/Switch:Tactile Switches	2	EN1,IO0	KEY-SMD_L6.0-W3.6-LS8.3-L		TC-3602-4.3-260G	0.369	TC-3602-4.3-260G	0.3688	TC-3602-4.3-260G	TC-3602-4.3-260G
20	磁珠/滤波器/EMI优化:磁珠	2	L1,L2	L0805		CBW201209U301T	0.109	CBW201209U301T	0.1092	CBW201209U301T	CBW201209U301T
21	电感/线圈/变压器:功率电感	1	L3	IND-SMD_L2.5-W2.0-VLS252012CX-150M	2.2uH	FXLB252012-2R2-M	0.472	2.2uH	0.4742	FXLB252012-2R2-M	2.2uH
22	磁珠/滤波器/EMI优化:磁珠	2	L4,L5	L0805		CBM201209U301T	0.105	磁珠	0.1045	CBM201209U301T	磁珠
23	电感/线圈/变压器:功率电感	4	L6,L7,L8,L12	IND-SMD_L4.6-W4.1-A	22uH	SRG0420-220-M	0.881	22uH	0.8807	SRG0420-220-M	22uH
24		1	L11	L0603	2nh			2nh		L_0603	2nh
25	Optoelectronics:LED Indication - Discrete	1	LED1	LED0603-RD_RED_2		NCD0603R1	0.0622	NCD0603R1	0.0622	NCD0603R1_C84263_1	NCD0603R1
26	光电器件:发光二极管	2	LED2,LED3	LED_0603				LED_0603-R		LED_0603-R	
27	光耦/LED/数码管/光电器件:发光二极管/LED	1	LED4	led0603-rd_red_1		NCD0603R1	0.0622	NCD0603R1	0.0622	NCD0603R1_C84263	NCD0603R1
28		1	OLED2	0.96OLED_4P		0.96OLED_4P_MODULE_JX	0.004	0.96OLED_4P_MODULE_JX		0.96OLED_4P_MODULE_JX	0.96OLED_4P_MODULE_JX
29	三极管/MOS管/晶体管:三极管(BJT)	2	Q1,Q2	SOT-23-3_L2.9-W1.3-P1.90-LS2.4-BR		SS8050	0.0662	SS8050	0.0669	SS8050_C7420370	SS8050
30	电阻:贴片电阻	2	R1,R3	R0603_4	100K			100K		Res_0603	100K

31	Resistors:Chip Resistor - Surface Mount	1	R2	R0603_1	2kΩ	0603WAF2001T5E	0.0085	2kΩ	0.0085	0603WAF2001T5E	
32		1	R4	R0603_4	680kΩ	0603WAF6803T5E	0.0093	680kΩ	0.0093	0603WAF6803T5E	680kΩ
33		1	R5	R0603_4	150kΩ	0603WAF1503T5E	0.0078	150kΩ	0.0078	0603WAF1503T5E	150kΩ
34		1	R6	R0603_4	10kΩ	0603WAF1002T5E	0.0079	10kΩ	0.0079	0603WAF1002T5E	10kΩ
35	Resistors:Chip Resistor - Surface Mount	2	R8,R9	R0603_5	5.1kΩ	RC0603FR-075K1L	0.0094	5.1kΩ	0.0094	RC0603FR-075K1L	5.1kΩ
36	电阻:贴片电阻	2	R10,R19	R0603_5	100kΩ	RC0603FR-07100KL	0.0084	100kΩ	0.0084	RC0603FR-07100KL	100kΩ
37	Resistors:Chip Resistor - Surface Mount	2	R11,R17	R0603_5	1kΩ	RC0603FR-071KL	0.0088	1kΩ	0.0088	RC0603FR-071KL	1kΩ
38	Resistors:Chip Resistor - Surface Mount	1	R12	R0603_5	1.2kΩ	RC0603FR-072K2L		1.2kΩ		RC0603FR-072K2L	1.2kΩ
39	Resistors:Chip Resistor - Surface Mount	3	R13,R14,R15	R0603_5	10kΩ	RC0603FR-0710KL	0.0089	10kΩ	0.0089	RC0603FR-0710KL	10kΩ
40	Resistors:Chip Resistor - Surface Mount	1	R16	R0603_5	500mΩ	RL0603FR-070R5L	0.118	500mΩ	0.1175	RL0603FR-070R5L	500mΩ
41	电阻:贴片电阻	10	R20,R21,R22,R23,R24,R25,R26,R27,R28,R29	R0603_4	10K			10K		Res_0603	10K
42	电阻:贴片电阻	1	R49	r0603_3	2.2kΩ	RS-03K222JT	0.0076	2.2kΩ	0.0076	RS-03K222JT	2.2kΩ
43	Resistors:Chip Resistor - Surface Mount	1	R50	R0603_1	226kΩ	0603WAF2263T5E	0.0078	226kΩ	0.0078	0603WAF2263T5E	
44		3	R51,R52,R53	R0603				10k		10k	10k
45	按键/开关:滑动开关	1	SW1	SW-TH_SK12D07VG5-1		SK12D07VG5	0.14	SK12D07VG5	0.1413	SK12D07VG5	SK12D07VG5
46	运算放大器/比较器:音频功率放大器	2	U1,U2	TQFN-16_L3.0-W3.0-P0.50-BL-EP1.5		MAX98357AETE+T	8.8	MAX98357AETE+T	8.8	MAX98357AETE+T	MAX98357AETE+T
47	连接器:线对板针座	3	U3,U23,U24	CONN-TH_2P-P1.25_MEGASTAR_ZX-MX1.25-2PZZ		ZX-MX1.25-2PZZ	0.0883	ZX-MX1.25-2PZZ	0.0892	ZX-MX1.25-2PZZ	ZX-MX1.25-2PZZ

48	电源管理:DC-DC电源芯片	1	U4	SOT-23-5_L3.0-W1.7-P0.95-LS2.8-BR_1		SY8088AAC	0.573	SY8088AAC	0.5728	SY8088AAC	SY8088AAC
49	电源芯片:电池管理	1	U5	ESOP-8_L4.9-W3.9-P1.27-LS6.0-BL-EP		TP4056	0.403	TP4056	0.4073	TP4056_C725790	TP4056
50		1	U6	SOT-23-3_L2.9-W1.6-P1.90-LS2.8-BR		AO3401	0.004	AO3401		AO3401	AO3401
51		1	U12	SW-SMD_EC1106S		EC1106S	0.004	EC1106S		EC1106S	EC1106S
52	电源管理:线性稳压器(LDO)	2	U15,U25	SOT-23-5_L3.0-W1.7-P0.95-LS2.8-BR		ME6217C33M5G	0.485	ME6217C33M5G	0.4848	ME6217C33M5G	ME6217C33M5G
53	通信接口芯片:USB转换芯片	1	U17	SOP-16_L10.0-W3.9-P1.27-LS6.0-BL		CH340C	3.6	CH340C	3.6	CH340C	CH340C
54	物联网/通信模块:WiFi模块	1	U21	WIFI-SMD_ESP32-WROOM-32E		ESP32-WROOM-32E-N4	23.46	ESP32-WROOM-32E-N4	23.46	ESP32-WROOM-32E-N4	ESP32-WROOM-32E-N4
55	Connectors:USB Connectors	1	USB1	USB-C-SMD_TYPE-C-6PIN-2MD-073		TYPE-C 16PIN 2MD(073)	0.409	TYPE-C 16PIN 2MD(073)	0.4127	TYPE-C 16PIN 2MD(073)	TYPE-C 16PIN 2MD(073)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